ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

# Weather Monitoring and Forecasting System

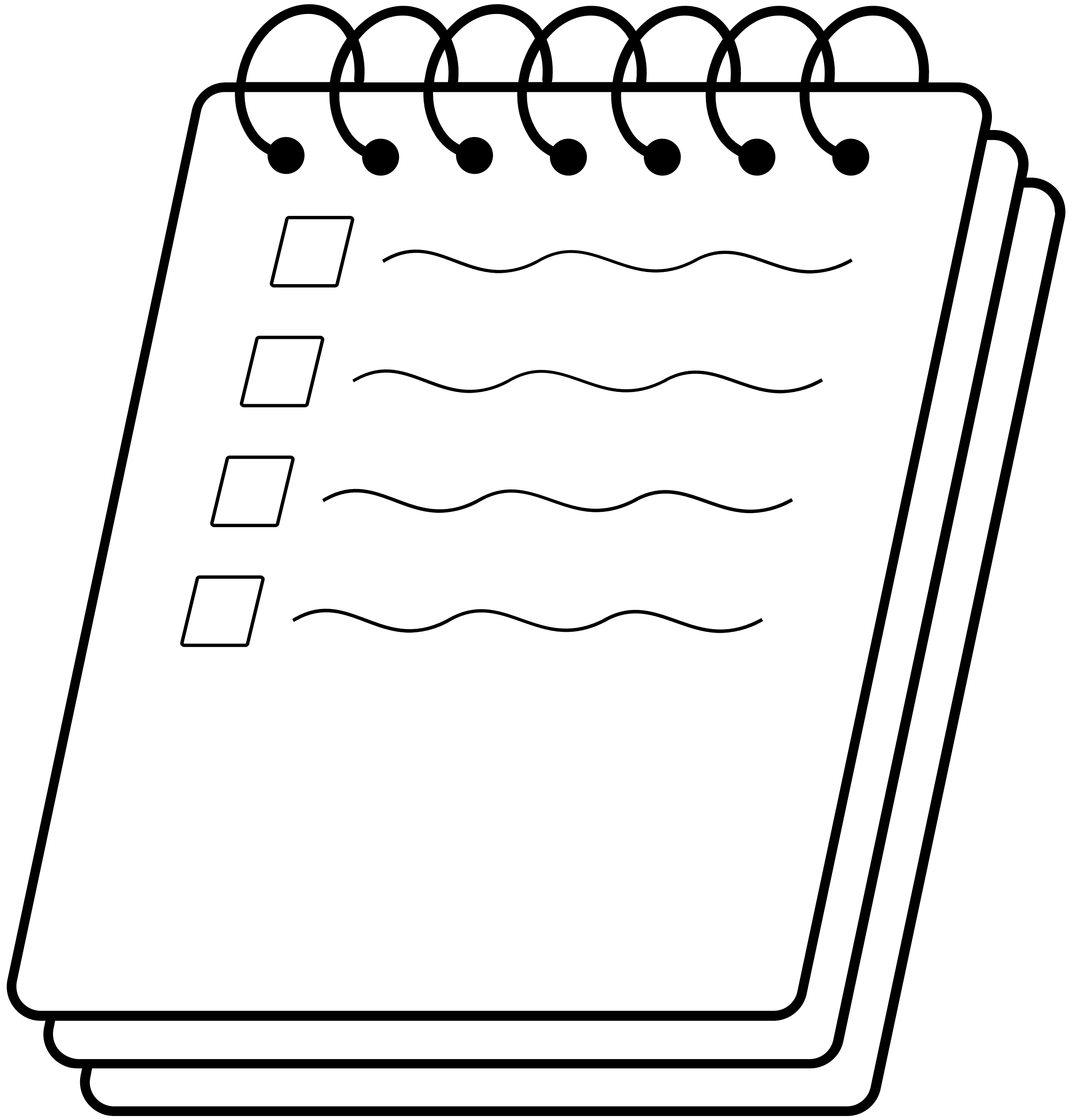
## Big Data Storage and Processing


Group 13

*Nguyễn Trí Thanh 20225457*
*Phan Trần Việt Bách 20225435*
*Lê Nhật Quang 20225522*
*Nguyễn Hoàng Sơn 20225525*
*Nguyễn Minh Quân 20225520*


ONE LOVE. ONE FUTURE.

# Table of contents



## Motivation

- Every year, the East Sea faces numerous tropical storms and typhoons, causing severe impacts on coastal and low-lying communities.
- Timely warnings can help fishermen, local authorities, and citizens prepare effectively and minimize damage.
- Flood and inundation alerts, providing early warnings for areas at high risk of water overflow and urban flooding, helping communities take preventive actionsbefore severe impacts occur.
- Our project, Weather Monitoring and Forecasting System using Big Data, aims to enhance the accuracy and speed of weather prediction by analyzing vast amounts of real-time meteorological and hydrological data from satellites, sensors, and IoT devices.

! URGENT

# 2. Dataset

## Leveraging Multiple Data Types

| Data Type | Description / Source | Common Fields | Main Contribution |
|---|---|---|---|
| **Rainfall Observation (Ground Stations)** | Rainfall measured by national meteorological & hydrological stations | - Station ID- Location (lat, lon, elevation)- Timestamp- Rainfall (mm)- Data status | **Monitoring & Validation** |
| **Satellite / Gridded Rainfall Data** | Rainfall estimated from satellites or climate models (e.g., TRMM, GPM, CHIRPS) | - Grid (lat, lon)- Time- Rainfall (mm/h)- Quality/confidence | **Prediction & Spatial Monitoring** |
| **Water Level / River Discharge Data** | River water level and discharge at hydrological stations | - Station ID- Location- Timestamp- Water level (m), discharge ($m^3/s$) | **Monitoring & Flood Warning** |
| **Soil Moisture Data** | Measured or estimated soil saturation levels | - Area/grid ID- Time- Moisture (%) by depth | **Prediction (Runoff Modeling)** |
| **Topographic / Elevation Data (DEM, LiDAR)** | Digital elevation model of terrain | - Grid (lat, lon)- Elevation (m)- Slope / aspect | **Flood Mapping & Modeling** |
| **Land Cover / Land Use Data** | Classification of land surfaces (forest, urban, agriculture, etc.) | - Grid/vector- Land type- Impervious surface ratio | **Prediction & Flood Simulation** |
| **Flood Inundation Maps (Remote Sensing)** | Flooded areas detected by SAR or optical satellite images | - Grid (lat, lon)- Time- Flood depth / binary- Confidence | **Flood Detection & Validation** |
| **Meteorological Parameters** | Temperature, pressure, humidity, wind, radiation, evaporation | - Location- Time- Variables: temp, pressure, humidity, wind, radiation, evapotranspiration | **Prediction (Weather Modeling)** |
| **Urban / Social Sensor Data** | IoT water sensors, CCTV, crowd reports, traffic flow | - Sensor/camera ID- Timestamp- Water depth / wetness / image | **Monitoring & Early Warning** |

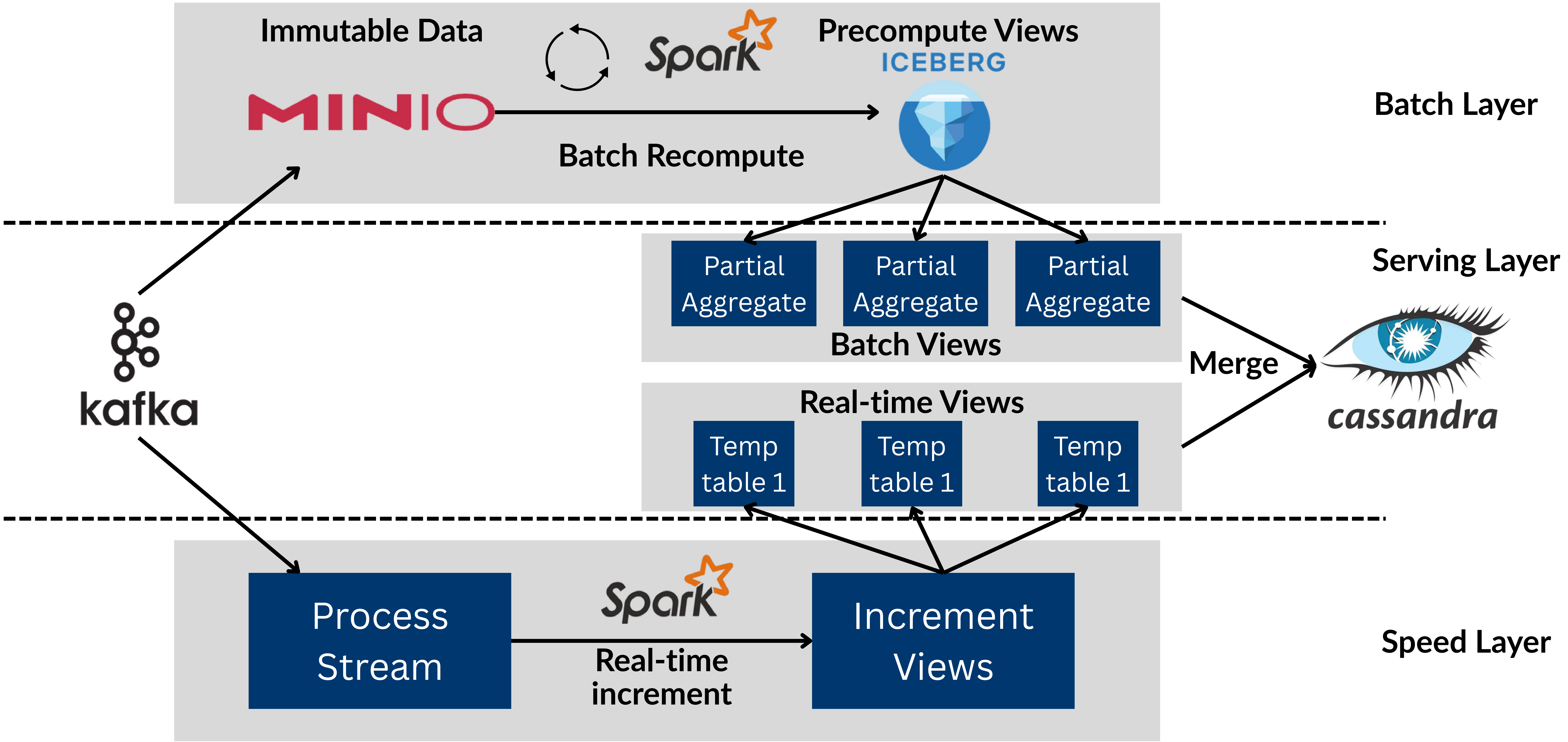
# 2. Dataset

## Integration of multiple data sources

- Rainfall & Weather Data: Rainfall observation, CHIRPS rainfall, GPM satellite rainfall, Open-Meteo weather & flood API, ...
- Hydrology & Flood Forecasting: GloFAS (Global Flood Awareness System), GFMS (Global Flood Monitoring System, NASA/UMD), DAHITI (Satellite Water Level Database), ...
- Remote Sensing & Flood Mapping: MODIS/VIIRS flood maps, Sentinel-1 SAR imagery, Global Flood Monitoring (Copernicus GFM), SRTM DEM (NASA/USGS), ESA WorldCover 2020 (10 m land cover)
- Meteorological & Environmental Data: NASA POWER climate & solar dataset, ERA5 reanalysis (Copernicus), Vietnam climate data 1901–2020
- Urban & Real-time Social Data: Vietnam IoT flood sensors (Smart City projects), Google Maps traffic data
- Vietnam & Regional Open Data: Open Development Vietnam portal, UN-SPIDER GIST datasets for Vietnam , VMHA / KISTERS National Flood Forecasting System

# 3. System Architecture

**Lambda Architecture:** for batch and stream processing



# 3. System Architecture

| Layer | Technology | Description |
|---|---|---|
| Message Queue | Apache Kafka | Collects and streams incoming data from various producers to downstream consumers in real time |
| Distributed Storage | MinIO | Stores raw input data files and holds Iceberg's metadata and processed data files |
| Table format & metadata management | Apache Iceberg | Provides batch views with schema evolution, ACID transactions, and version control via snapshots |
| Batch Processing | Apache Spark | Performs ETL on raw data and forms batch views |
| Stream Processing | Spark Structured Streaming | Consumes Kafka streams to produce real-time views with incremental updates |
| Serving layer | Cassandra | NoSQL database that stores merged batch and stream views, enabling fast, low-latency queries |

HUST

# THANK YOU !